

Climate policies exclusively centered in the Global North harm the economy of the Global South

Policy lessons of a two-region input-output ecological macroeconomic model

Oriol Vallès Codina

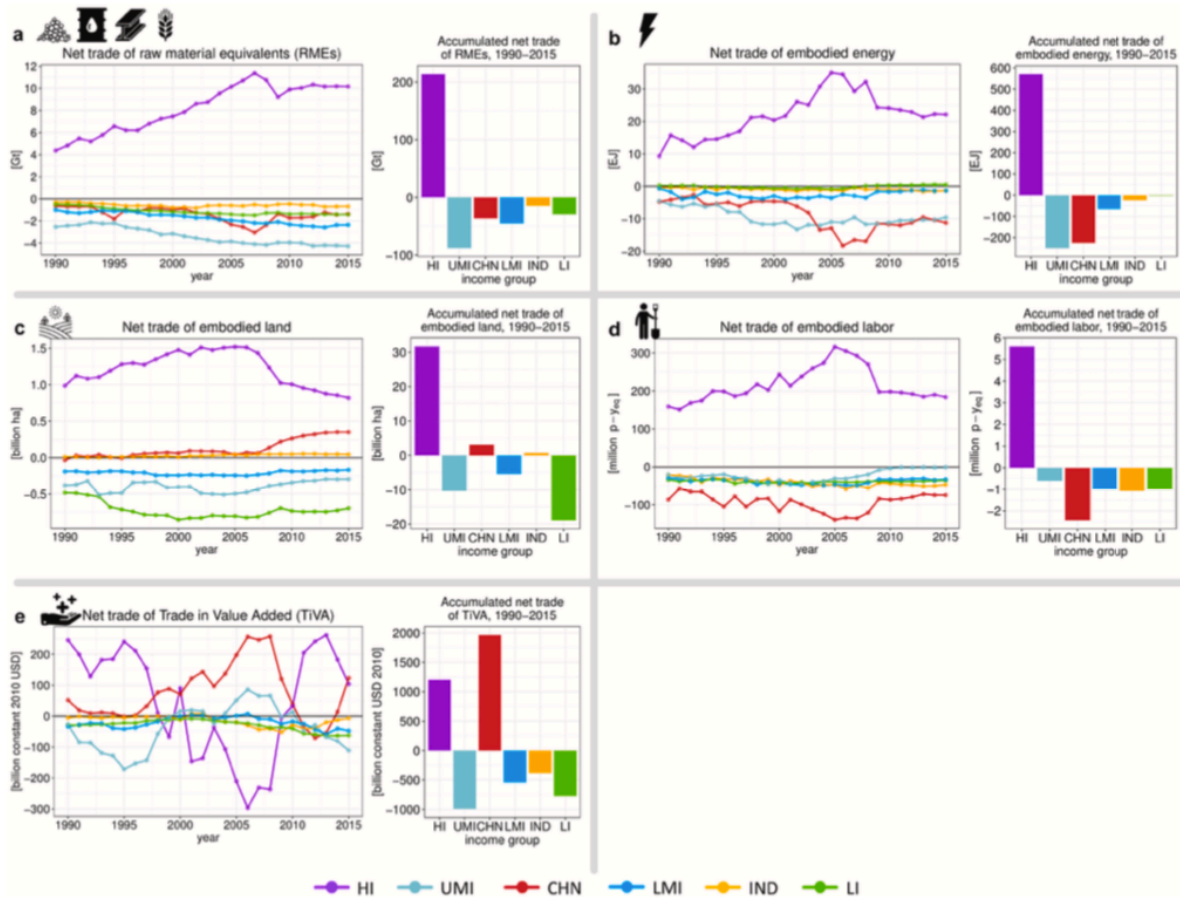
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Introduction

Outline

- ECO-MRIO-SFC modeling as a tool to understand the very **complex impact of (carbon) taxes and tariffs on global value chains** from a socio-economic perspective: ecological, macroeconomic and distributional implications for class, gender, race / region
 - Trade wars and global value chain fragmentation/regionalization
 - **Challenge** how to overcome ecological unequal exchange?
 - Current model shows ecological and economic impacts are best in cooperative models
 - Implications: competition vs cooperation in trade (Blecker 2025)
-

Motivation: Exchange is Unequal



Starting Point: the JUST2CE project

- **JUST2CE** = A Just Transition to the Circular Economy
- EU-funded project proposing an alternative way of looking at CE
- Most projects have focused on *how* to produce circularity. JUST2CE focuses on *what* (democracy, participation, gender, global justice)
- Two milestones / WP5 deliverables:

- **Systematic review** of macro models assessing CE transition (**Feverheiro et al 2025**)
 - **Formal model(s)** to simulate & compare CE policies and transition scenarios (**Vallès Codina et al., 2026?**)
-



Macroeconomic models for assessing the transition towards a circular economy: A systematic review

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Marco Veronese Passarella ^{d e}

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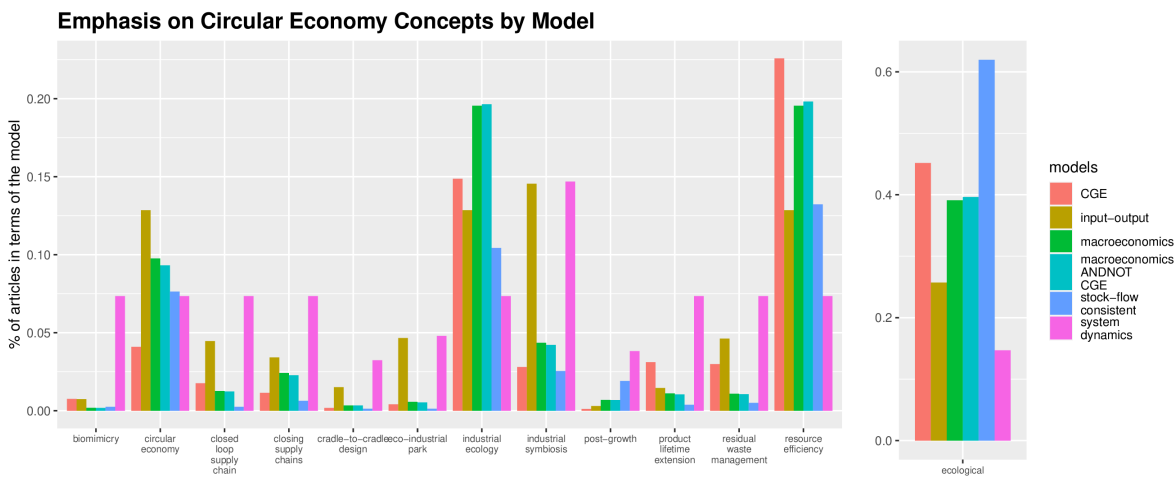
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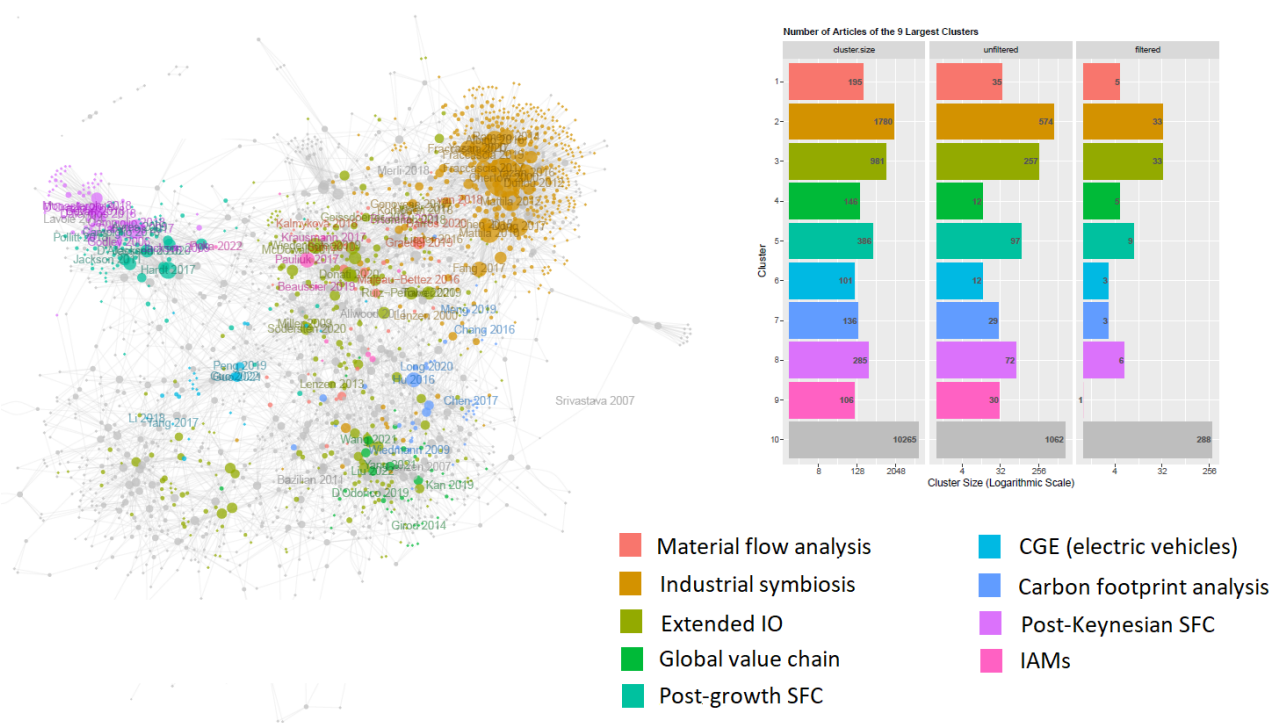
Abstract

The Circular Economy (CE) paradigm has gained traction in both academic discourse and industrial practice. While a transition towards a CE is generally associated with more sustainable futures, less is known about its socio-economic feasibility. This article provides a systematic literature review of contributions to macroeconomic modelling which evaluate environmental and socio-economic impacts of CE interventions (classified in terms of closing supply chains, resource efficiency, residual waste management, and product lifetime extension). Differences in modelling approaches (Leontief input-output, macroeconometric input-output, and computable general

Systematic Literature Review (Simplified)



Citation Network of Ecological Macroeconomics



Main Gaps Identified

- **Modelling of Rebound Effects** — limited treatment of demand/impact changes via prices, jobs, income
- **Transitional Dynamics** — IO analysis typically static
- **Limited Socio-Economic Coverage** — often only employment considered
- **Technology Innovation & Diffusion; Demand Assumptions** — low-labor-cost tech may be preferred
- **North–South / Core–Periphery Ecological Unequal Exchange** (Dorninger 2021, Hickel)

ECO–MRIO–SFC Modeling

The Model: Basic Features

Main tools in (theory-to-data) **macroeconomics**:

- 1) **DSGE (and growth) models** — *Pros*: mainstream, simple story. *Cons*: “small”, unique optimal equilibrium
- 2) **CGE models (and IAMs)** — *Pros*: many variables, cross-industry links. *Cons*: unique optimal equilibrium, no finance

Cross-breeding two alternatives (*Hardt & O’Neill 2017*):

- **Leontief IO Models** — *Pros*: many variables, cross-industry links. *Cons*: static, no finance
 - **SFC Models** — *Pros*: dynamics, finance. *Cons*: homogeneous output
-

Theoretical Antecedents

- Increasing use of IO–SFC models; yet none focuses on CE:
 - *Berg, Hartley & Richters (2015)*. A stock-flow consistent input-output model with applications to energy price shocks, interest rates, and heat emissions.
 - **Other contributions** *Naqvi (2015)*; *Naqvi and Stockhammer (2018)*; Valdecantos & Valentini (2017); Nieto et al (2019); Jackson & Jackson (2021); Cordier et al. (2015); Jackson et al. (2014)
 - MEDEAS, Eurogreen, REAL, ToBe... increasingly very sophisticated one-region models
-

ECO–MRIO–SFC Model

- a) Two-country macro frame from standard SFC models (*Godley & Lavoie 2007*):
- **Six sectors**: households, production firms, government, commercial banks, central bank
 - **Four assets**: cash, bank deposits, shares, government bills (+ advances)
 - Only **loans to firms** (no personal loans)
 - **Fixed capital**, no inventories
- b) Flexible IO structure: **currently 50+ industries**
- c) Identification: calibration to target final demand components
- d) Solution: numerical simulations (R), 100 periods, 100 iterations
-

Final Demand and Gross Output: Constant Sector Shares

Let us consider a 3×3 production. The **final demand** vector is:

$$\mathbf{d} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} c + \begin{bmatrix} \iota_1 \\ \iota_2 \\ \iota_3 \end{bmatrix} i_d + \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix} gov + \begin{bmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{bmatrix} exp$$

The **gross output** vector is:

$$\mathbf{x} = \mathbf{Ax} + \mathbf{d} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{d}, \quad \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Classical Approach to Prices

For N regions, K industries per region, and $i = 1, \dots, KN$ industries:

$$p_{i,t} = (1 + \mu) \left[wl_i + \sum_{j=1}^{KN} p_{j,t-1} (A_{ji} + \delta_j K_{ji}) \right]$$

Prices are set sequentially as a markup over realised unit costs; under stationarity, the price vector converges to the fixed point of the corresponding IO price operator.

Carbon Taxation

Carbon taxes are internalized as a production cost for all commodities:

$$p' = p(1 + \tau)$$

where $p = wB$ is the price before tax, p' is the price with tax, and τ is the product of emission intensity times the carbon price. In scalar notation, we write $p_i = w \sum_j l_j b_{ji}$ so that adding the carbon price yields:

$$p'_i = w(1 + \tau_i) \sum_j l_j b_{ji}$$

(Carbon) Tariffs 1

We need to exploit the $KN \times KN$ dimensionality of global matrix of input-output coefficients A and thus the augmented Leontief matrix B . For $K = N = 2$, 2 products and 2 regions, labeled 1 to 4:

$$p = w(l_1, l_2, l_3, l_4) \begin{pmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \\ b_{41} & b_{42} & b_{43} & b_{44} \end{pmatrix}$$

(Carbon) Tariffs 2

In the carbon taxation case, we defined $1 + \tau$ as a KN -vector, but it can also be thought of a diagonal matrix with zeros in the off-diagonal. For the tariffs case, operate precisely on the off-diagonal matrices that refer to interregional trade. If region 1 imposes a tariff (τ_3, τ_4) on products 1 and 2 of region 2, we multiply coefficients b_{ij} by $(1 + \tau_i)$ for $i = 3, 4$ and $j = 1, 2$:

$$p' = w(l_1, l_2, l_3, l_4) \begin{pmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31}(1 + \tau_3) & b_{32}(1 + \tau_3) & b_{33} & b_{34} \\ b_{41}(1 + \tau_4) & b_{42}(1 + \tau_4) & b_{43} & b_{44} \end{pmatrix}$$

Ecological Unequal Exchange: EU (red) vs RoW (green)

Table of Industries

#

Industry

1

Agriculture

2

Animal Farming

3

Forestry, logging and related service activities (02)

4

Fishing, operating of fish hatcheries and fish farms; service activities incidental to fishing (05)

5

Extraction of Fossil-Fuels

6

Mining & Quarrying of raw materials

7

Processing of meat

8

Processing of other food & beverage

9

Manufacture of tobacco products (16)

10

Manufacture of textiles

11

Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials (20)

12

Re-processing of secondary wood material into new wood material

13

Pulp

14

Re-processing of secondary paper into new pulp

15

Paper

16

Processing of Fossil Fuels

17	
Plastics, basic	
18	
Re-processing of secondary plastic into new plastic	
19	
Manufacture of Chemicals	
20	
Manufacture of rubber and plastic products (25)	
21	
Manufacture of glass and glass products	
22	
Re-processing of secondary glass into new glass	
23	
Manufacture of other non-metallic mineral products	
24	
Manufacture of cement, lime and plaster	
25	
Re-processing of ash into clinker	
26	
Manufacture of Metals	
27	
Re-processing of Metals	
28	
Manufacture of durable goods	
29	
Manufacture of furniture; manufacturing n.e.c. (36)	
30	
Recycling	
31	

Production of electricity by fossil fuels

32

Production of electricity by renewable energy

33

Transmission & Distribution

34

Manufacture of gas; distribution of gaseous fuels through mains

35

Water

36

Construction (45)

37

Re-processing of secondary construction material into aggregates

38

Sale, maintenance, repair of motor vehicles, motor vehicle parts, motorcycles, motorcycle parts and accessories

39

Retail sale of automotive fuel

40

Retail & Wholesale trade

41

Leisure Services

42

Transport

43

KIBS

44

Real estate activities (70)

45

Renting of machinery and equipment without operator and of personal and household goods
(71)

46

Research and development (73)

47

Public administration and defence; compulsory social security (75)

48

Education (80)

49

Health and social work (85)

50

Incineration of Waste

51

Biogas & Composting

52

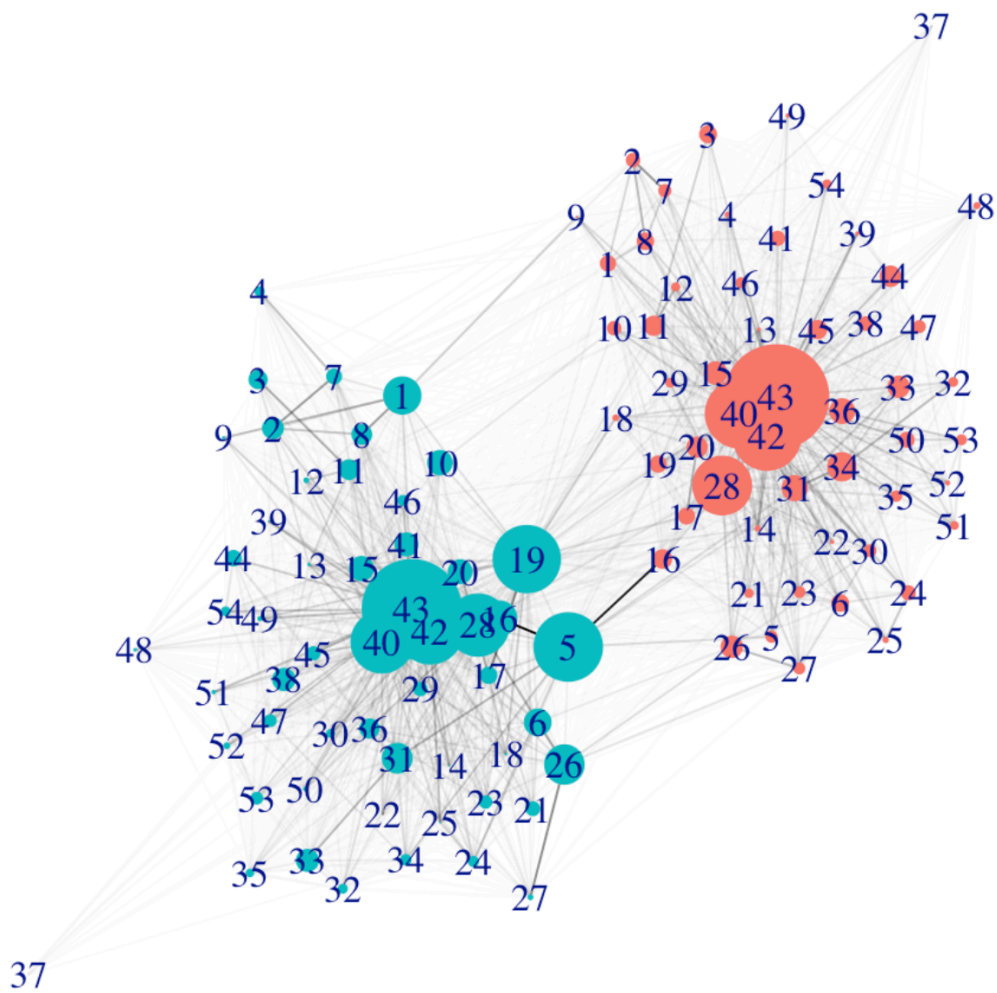
Waste water

53

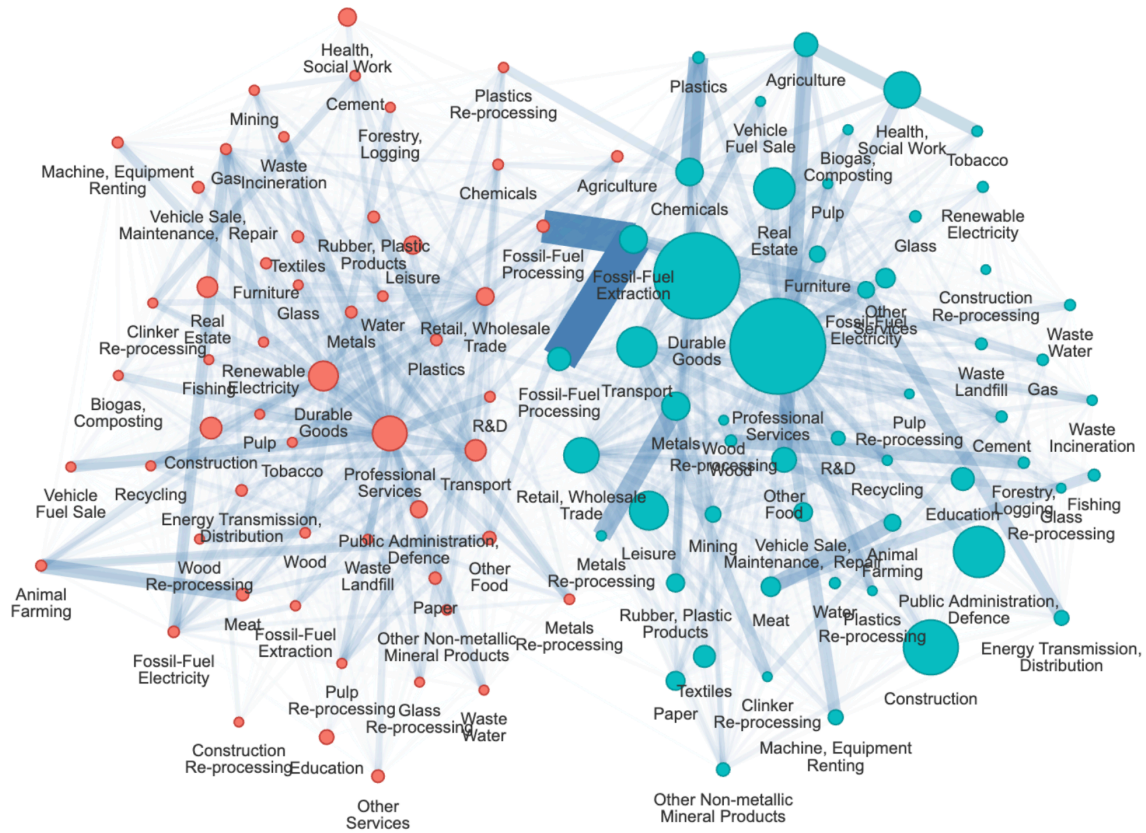
Landfill of Waste

54

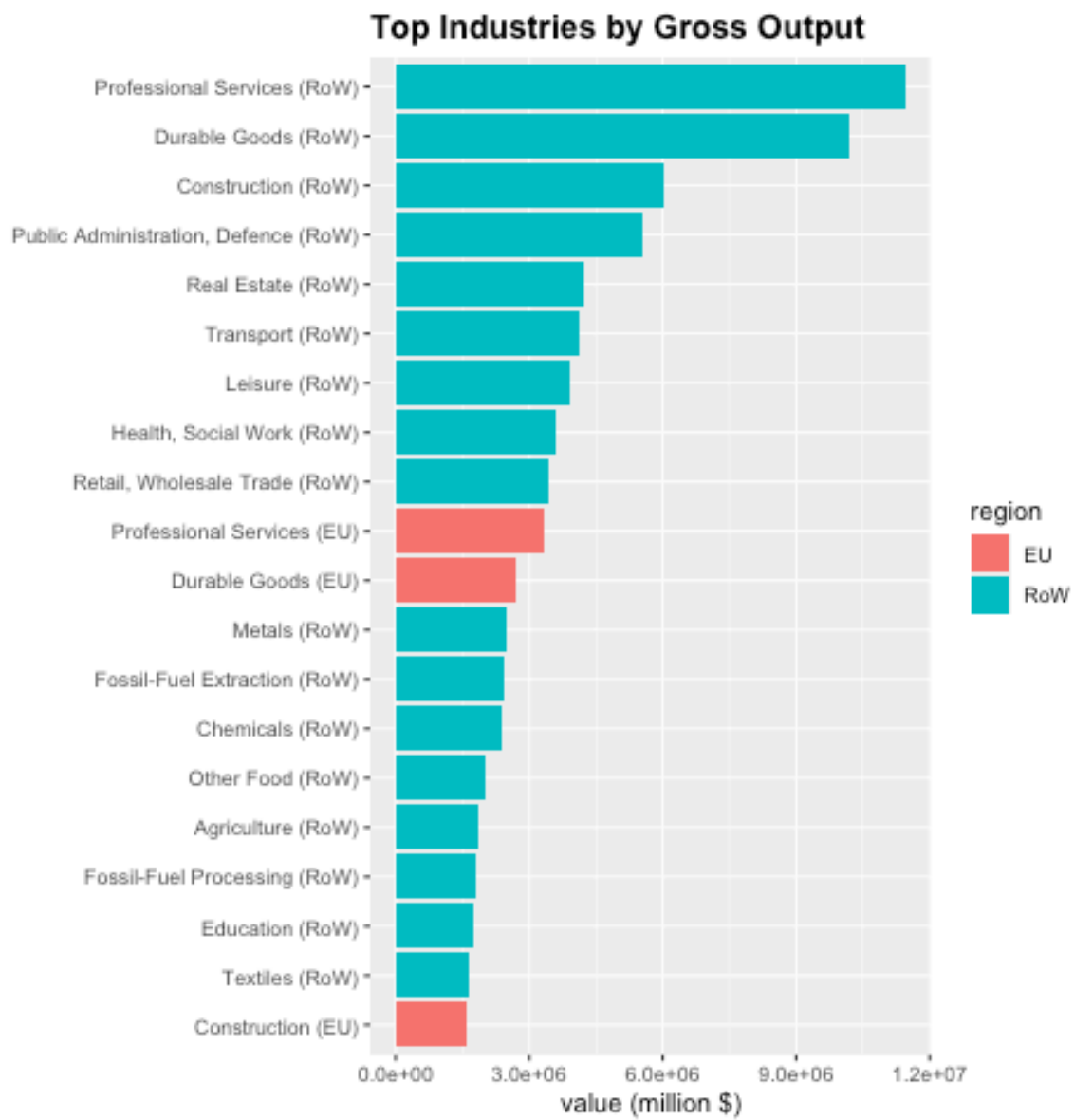
Other Services



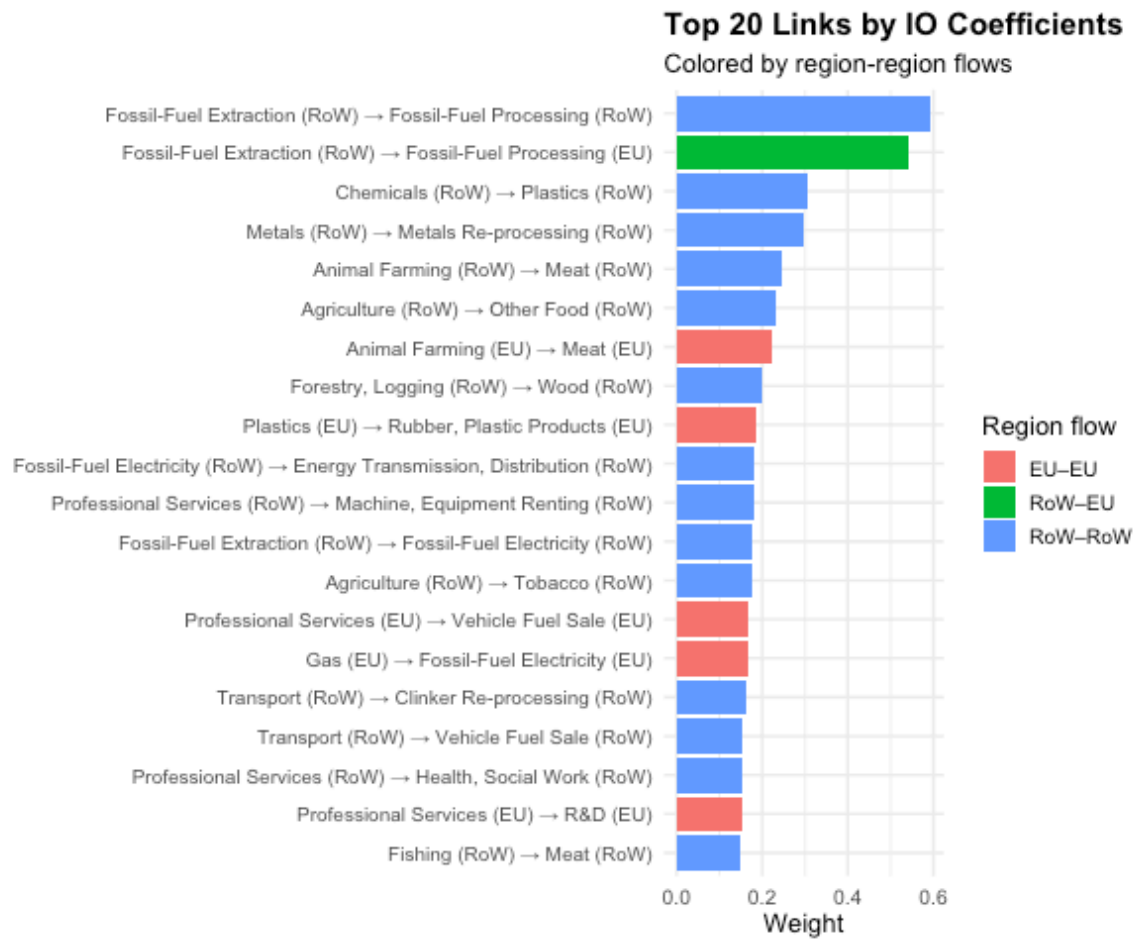
Input-Output Network



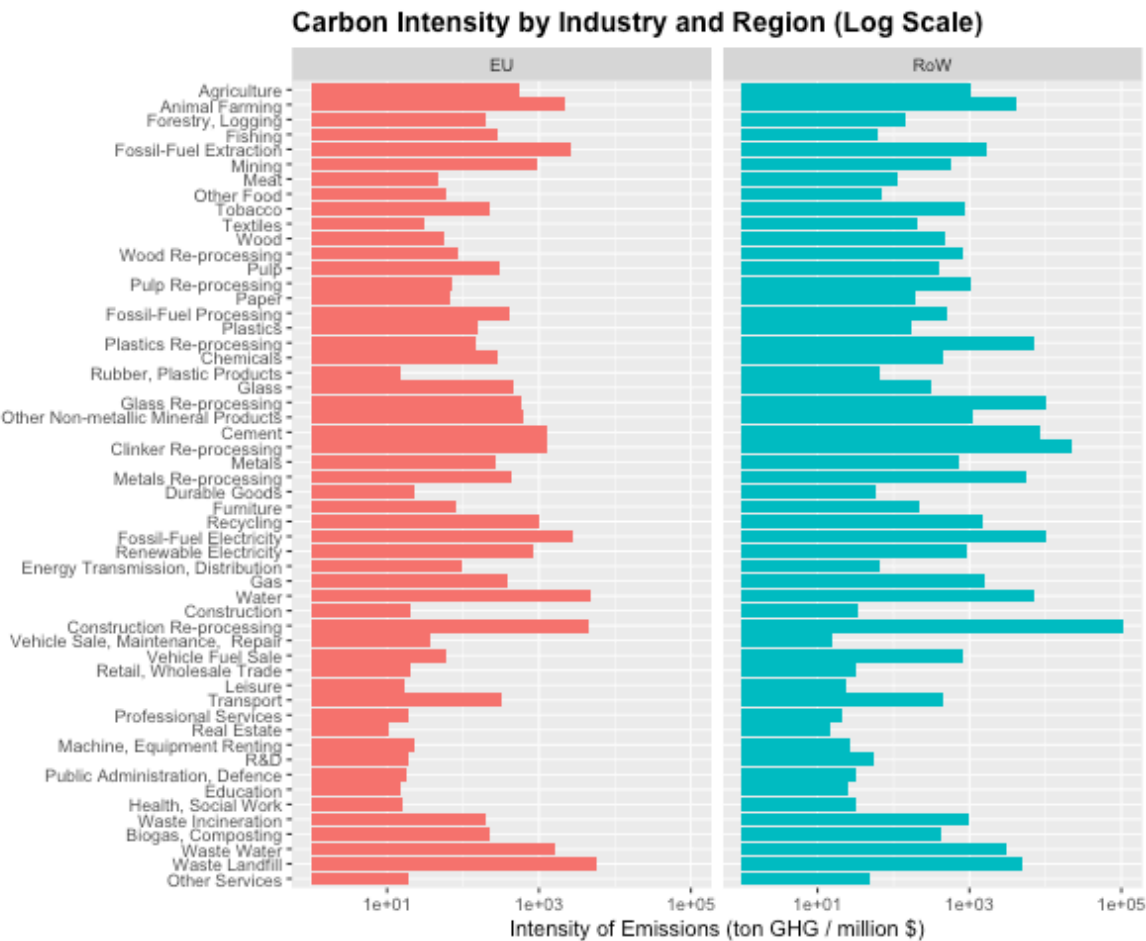
Top Nodes



Top Links

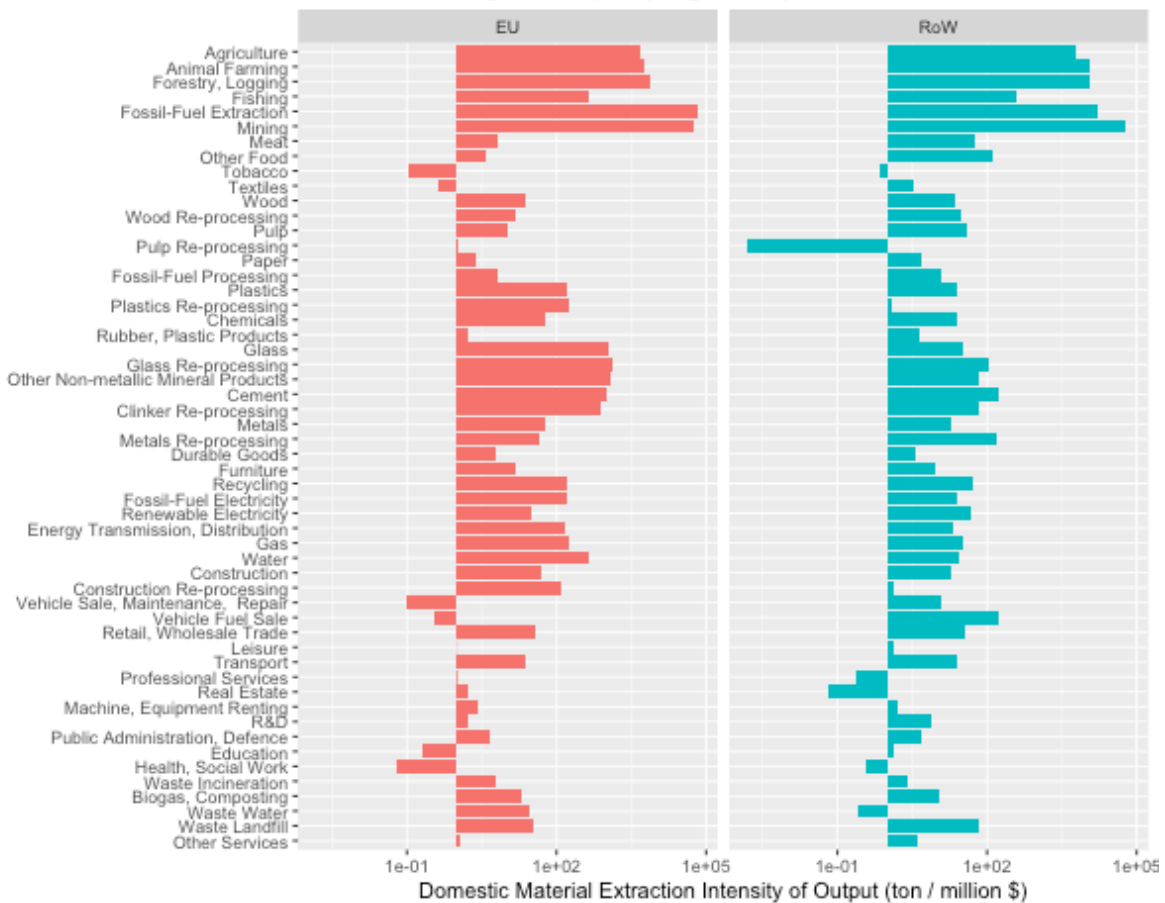


Carbon Intensity



Matter Intensity

Matter Intensity of Output (Log Scale)



Modeling Circular Economy Interventions 1: Product Lifetime Extension

Reuse and remanufacturing (a)

Industry							Consumers						
A	fp	c	pm	sm	sr	Y	A	fp	c	pm	sm	sr	Y
fp	-						fp						-
c	+						c	+					
pm							pm						
sm							sm						
sr	+						sr	+					+
b						b_y	b						b_y

A = Industry transactions
Y = Final demand
b = intermediate extensions
 b_y = final demand extensions

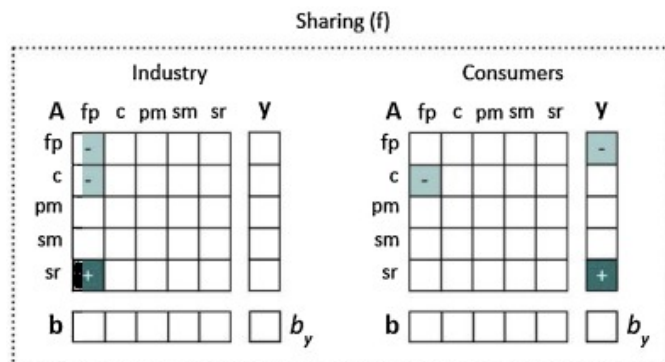
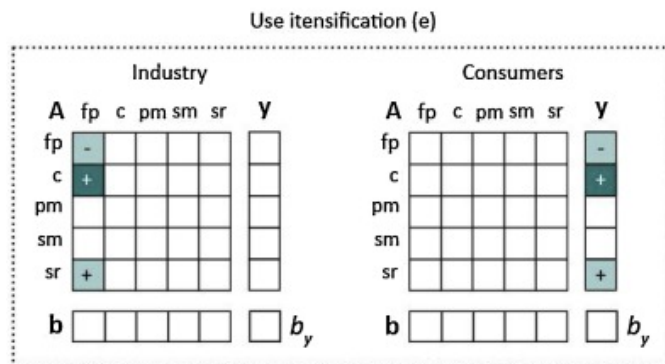
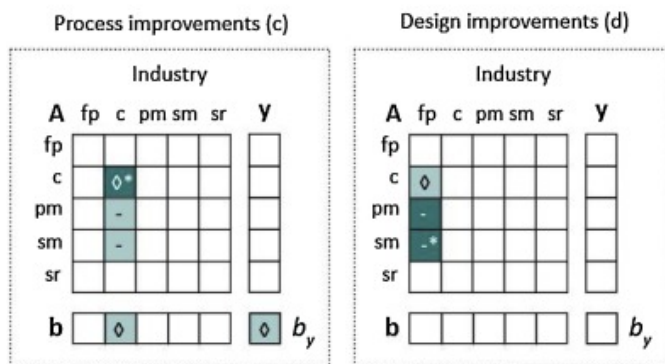
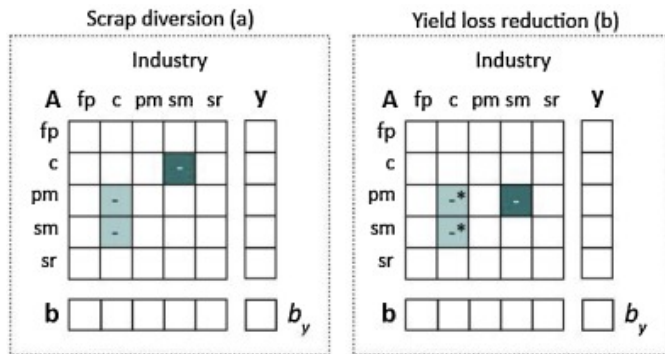
Delayed product replacement (b)

Industry							Consumers						
A	fp	c	pm	sm	sr	Y	A	fp	c	pm	sm	sr	Y
fp	-						fp						-
c							c						
pm	+						pm	+					
sm	+						sm	+					
sr	+						sr	+					+
b						b_y	b						b_y

fp = final manufactured product
c = component
pm = primary material
sm = secondary (recycled) material
sr = services

- = reduction
+ = increase
◊ = unclear change
* = substitution may apply
■ = primary change
■ = potential ancillary change

Modeling Circular Economy Interventions 2: Resource Efficiency



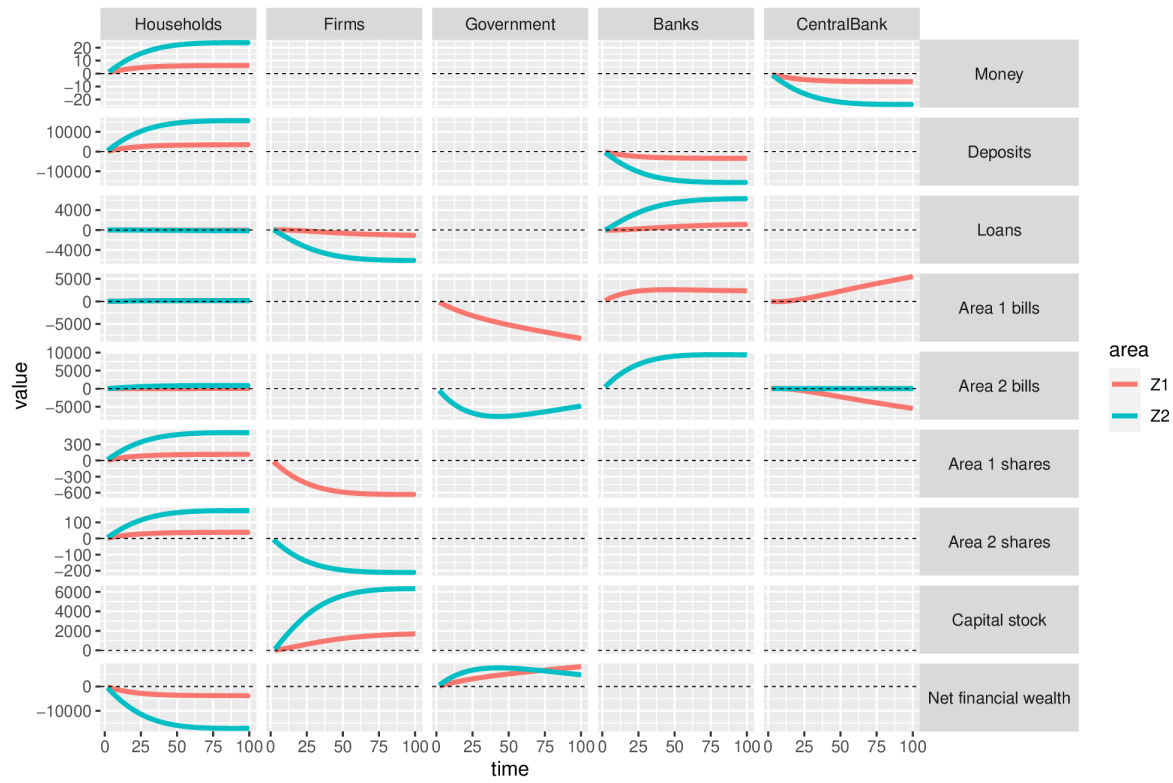
A = Industry transactions
 y = Final demand
 b = intermediate extensions
 b_y = final demand extensions

fp = final manufactured product
 c = component
 pm = primary material
 sm = secondary (recycled) material
 sr = services

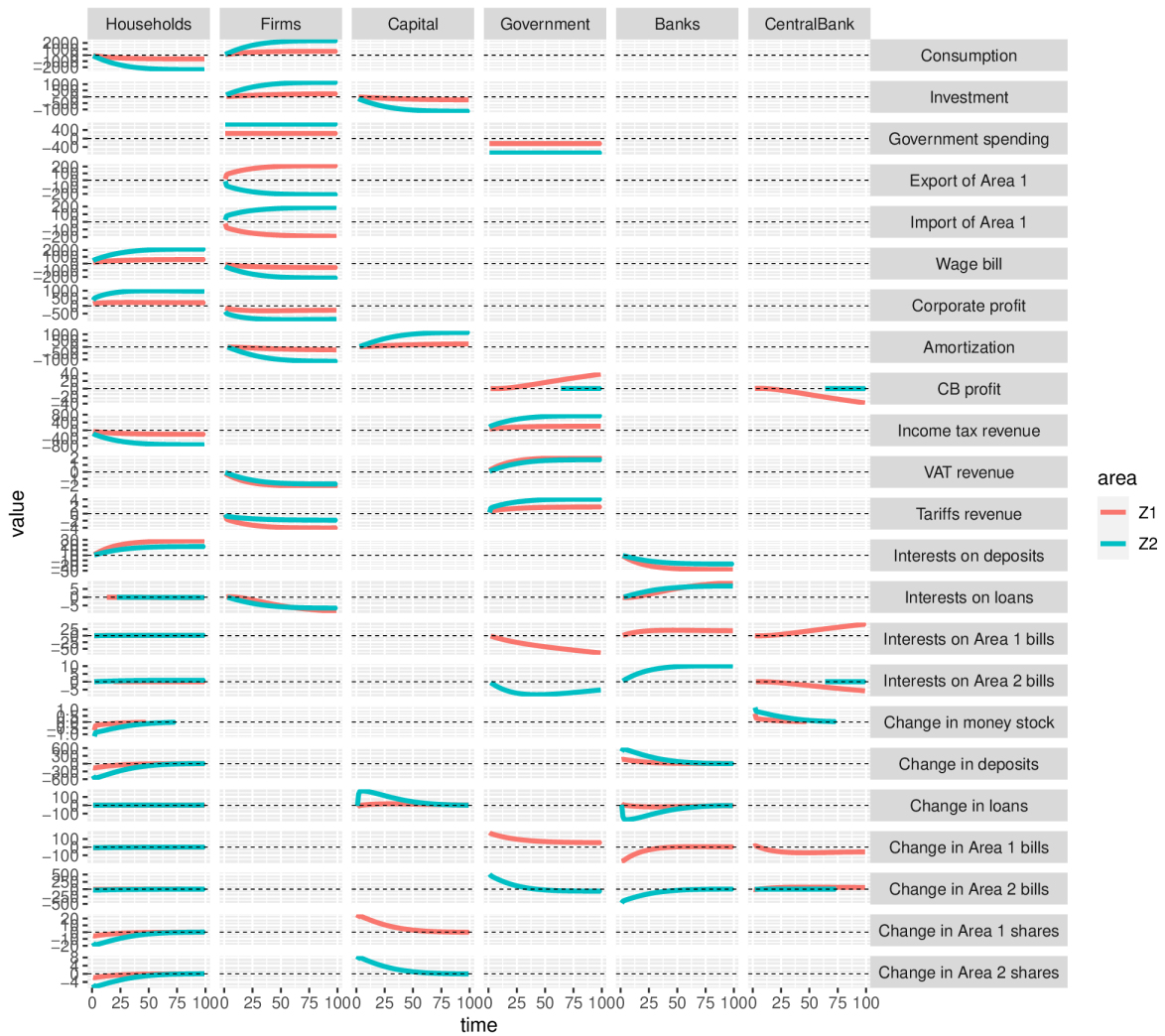
- = reduction
 + = increase
 ◇ = unclear change
 * = substitution may apply
 ■ = primary change
 ■ = potential ancillary change

Results

Balance Sheet



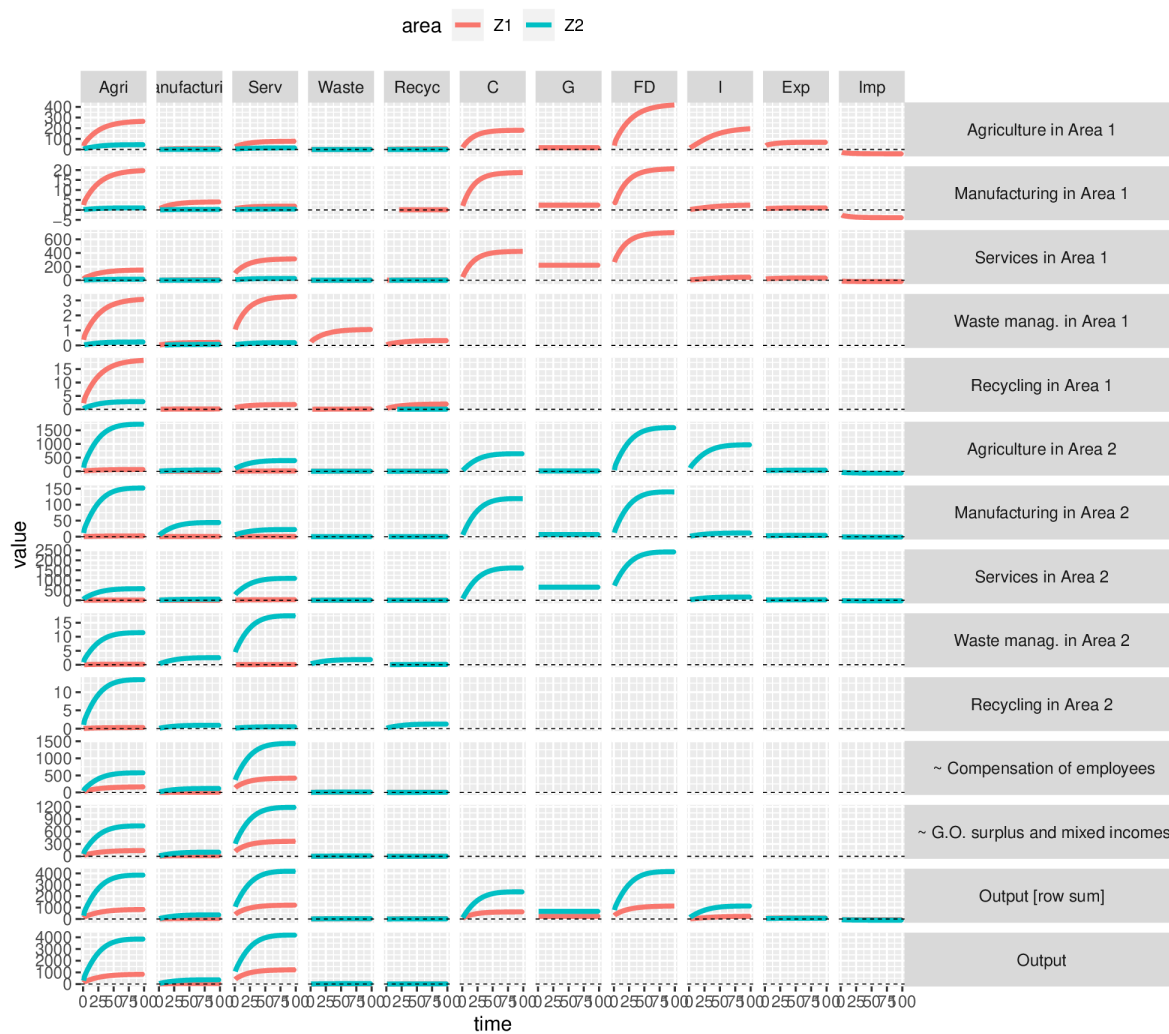
Transaction-Flow Matrix



TFM Formulas

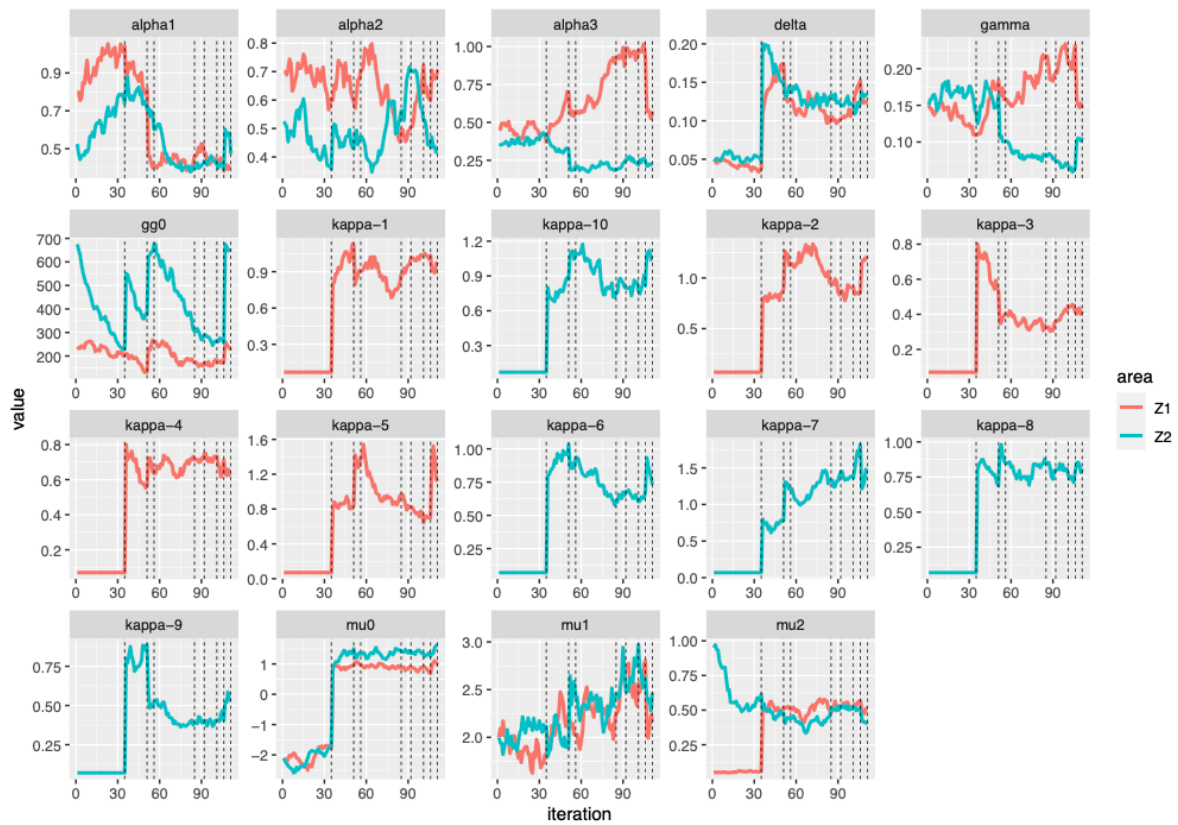
transaction	Firms_Current	Banks_Current	Central_Bank	Government
Consumption	- tfm Consumption Households + tfm ConsumptionofBankingservices Households + tfm PublicServices Households			
Public Services	- sna P.13 S.13 R/P + sna P.9 S.13 U/A			+ sna P.11 S.13 R/P - tfm PublicServices FirmsCurrent
Consumption of Banking services		- sna D.71 S.12 U/A + sna D.71 S.12 R/P - sna D.72 S.12 U/A + sna D.72 S.12 R/P - tfm ConsumptionofBankingservices Households	- sna D.71 S.12 U/A + sna D.71 S.12 R/P - sna D.72 S.12 U/A + sna D.72 S.12 R/P	
Investment	- tfm Investment FirmsCap - tfm Investment BanksCurrent - tfm Investment Government - tfm Investment Households	- sna P.5b S.12 U/A	- sna P.5b U/A	- sna P.5b S.13 U/A
Exports	- tfm Exports RoW			
Imports	- tfm Imports RoW			

Multi-Region Input–Output

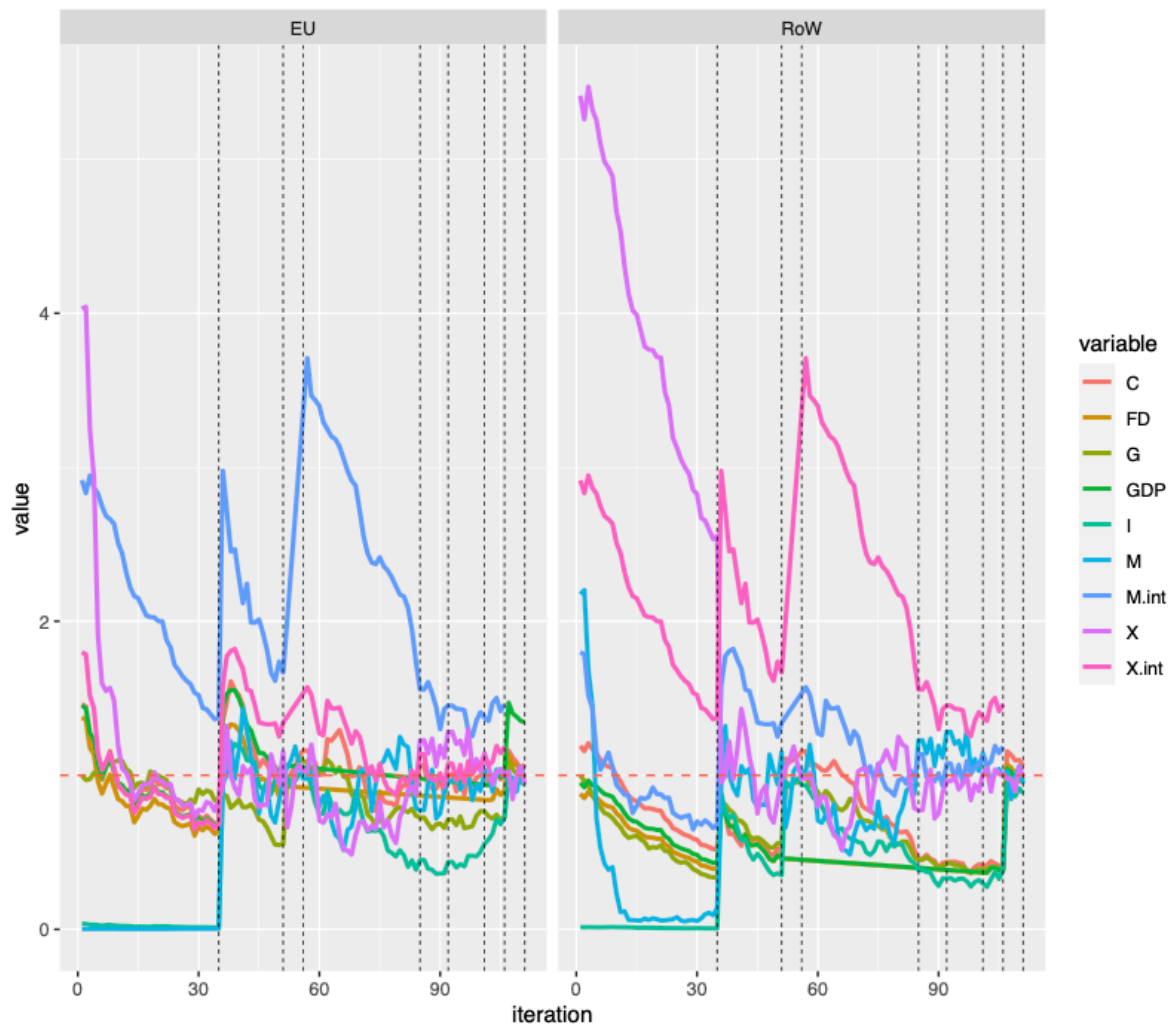


Calibration

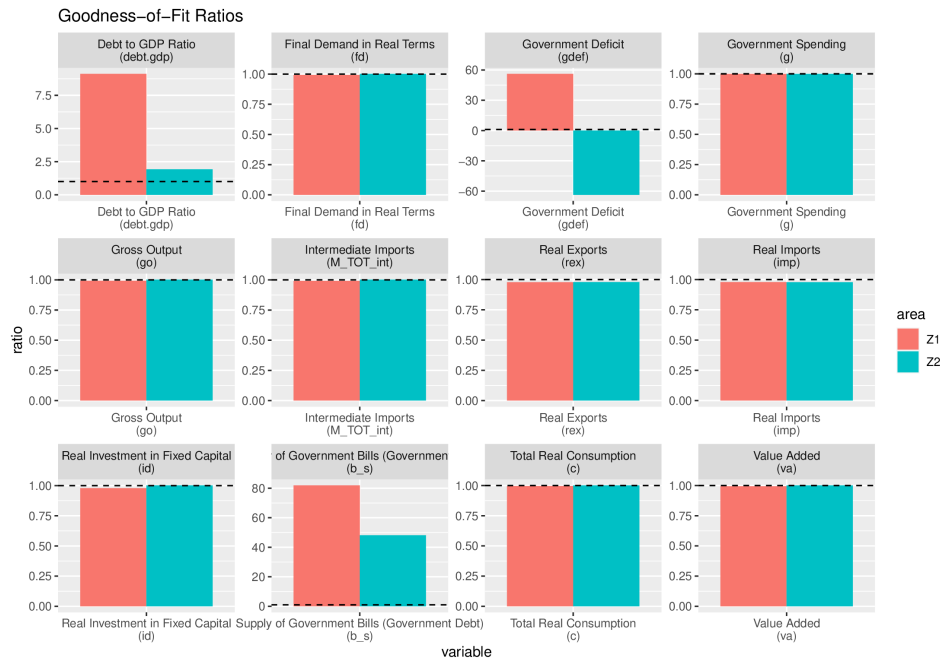
Instruments Evolution



Ratio Evolution



Calibration Result



Scenario Analysis

Shocks

Shock

Reduction in Consumption Level

Change in Consumption Composition towards Services

Product Life Time Extension

Higher Recycling Rate

Higher Propensity to Consume Green

Lower Extraction (or Conversion) Rate of Matter

Lower Discarding Rate of Socio-Economic Stock

Higher Renewable Energy Share

Higher Govt Spending towards Efficiency

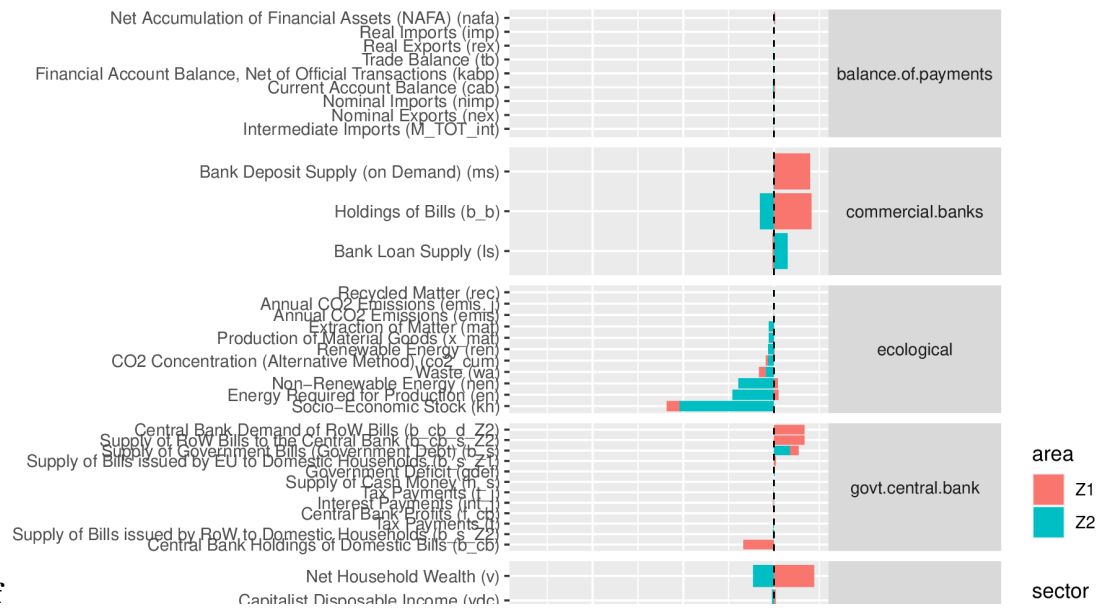
More Selective Govt Spending towards Recycling Efficiency

More Progressive Taxation

Scenario: Reduction in Consumption Level

Shock: Reduction in Consumption Level

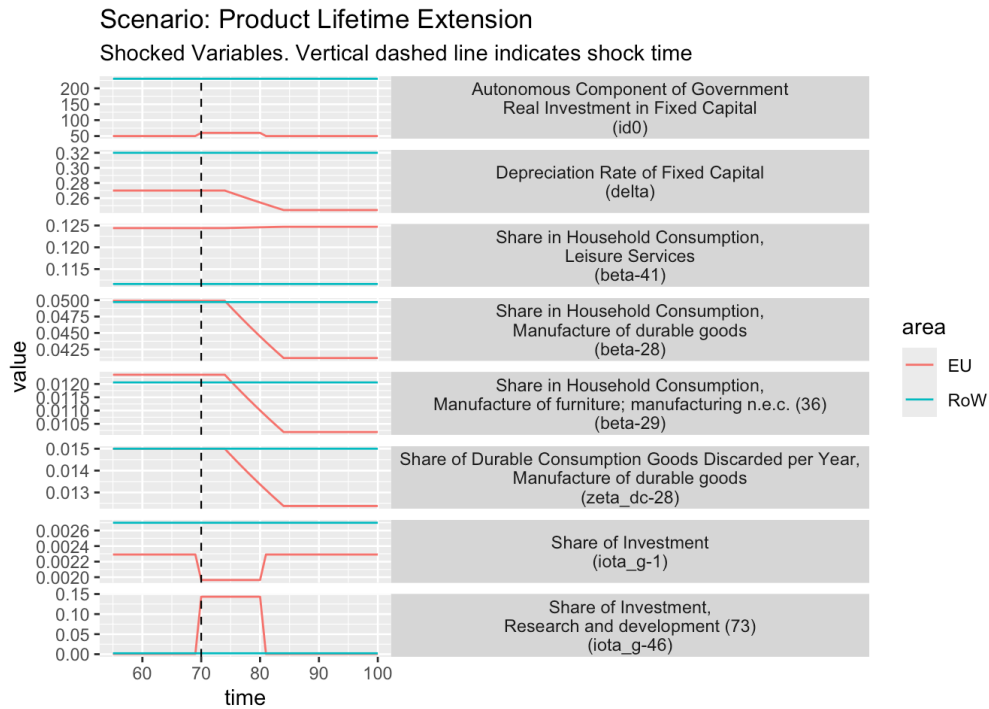
Shock–Baseline Difference, Last Period. Only Absolute Values Larger than



Top half



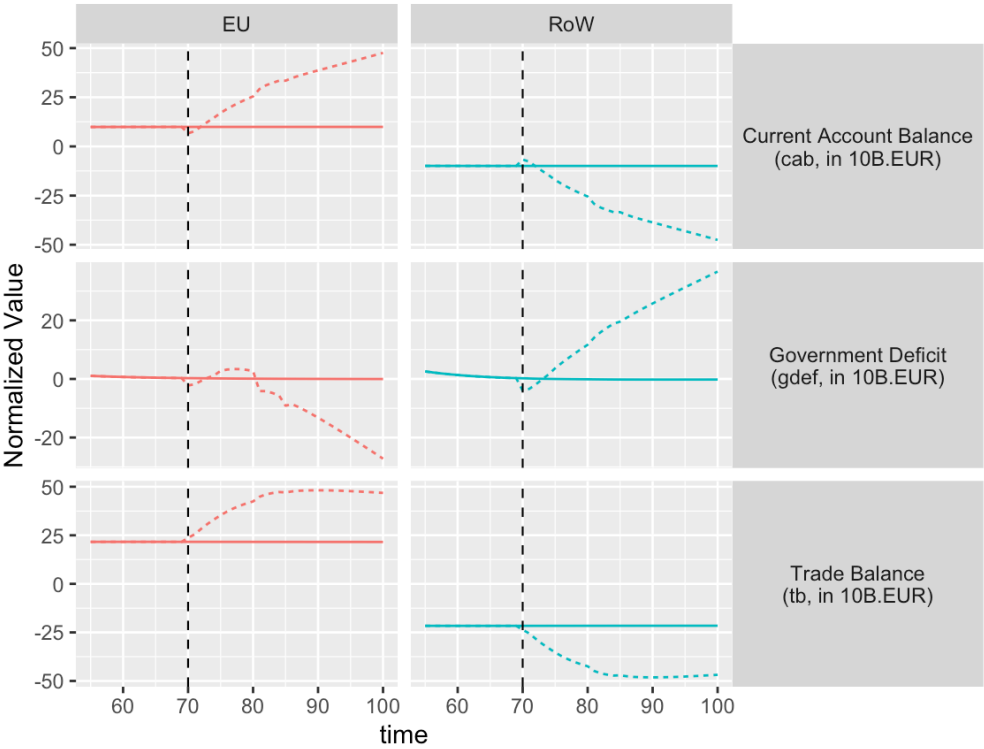
Scenario: Product Lifetime Extension



Impact: Product Lifetime Extension

Scenario: Product Lifetime Extension

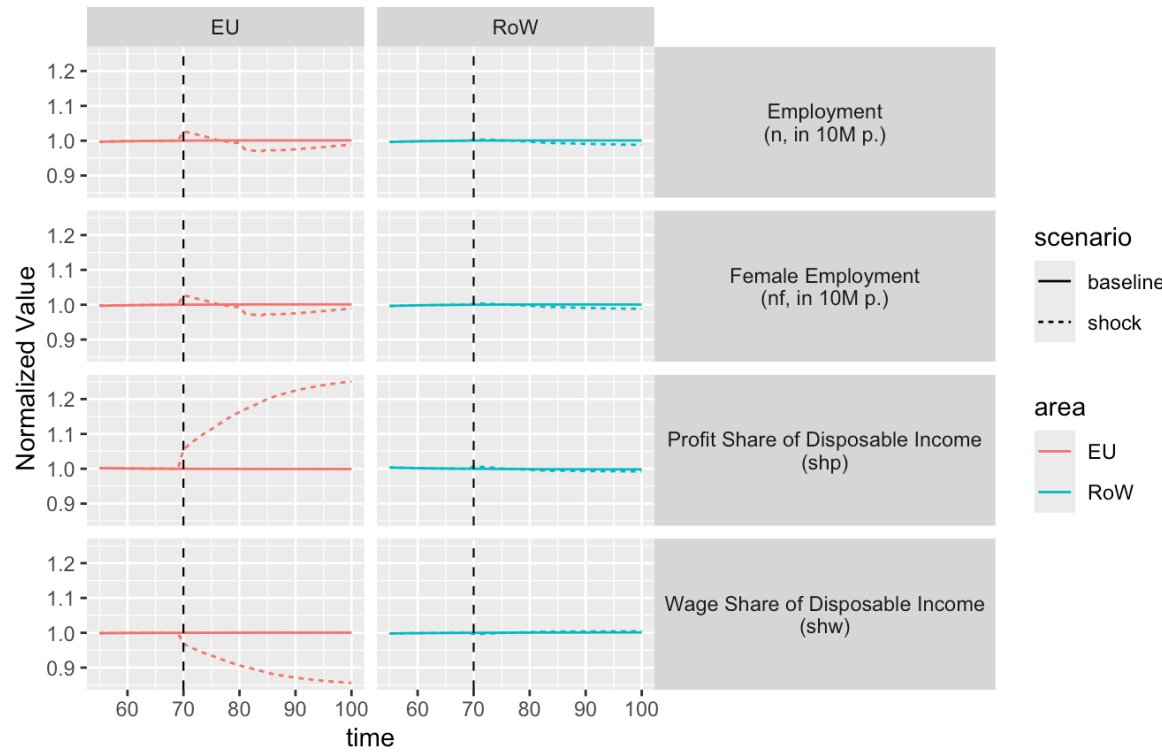
Selected External Indicators. Vertical dashed line indicates shock time



External Indicators

Scenario: Product Lifetime Extension

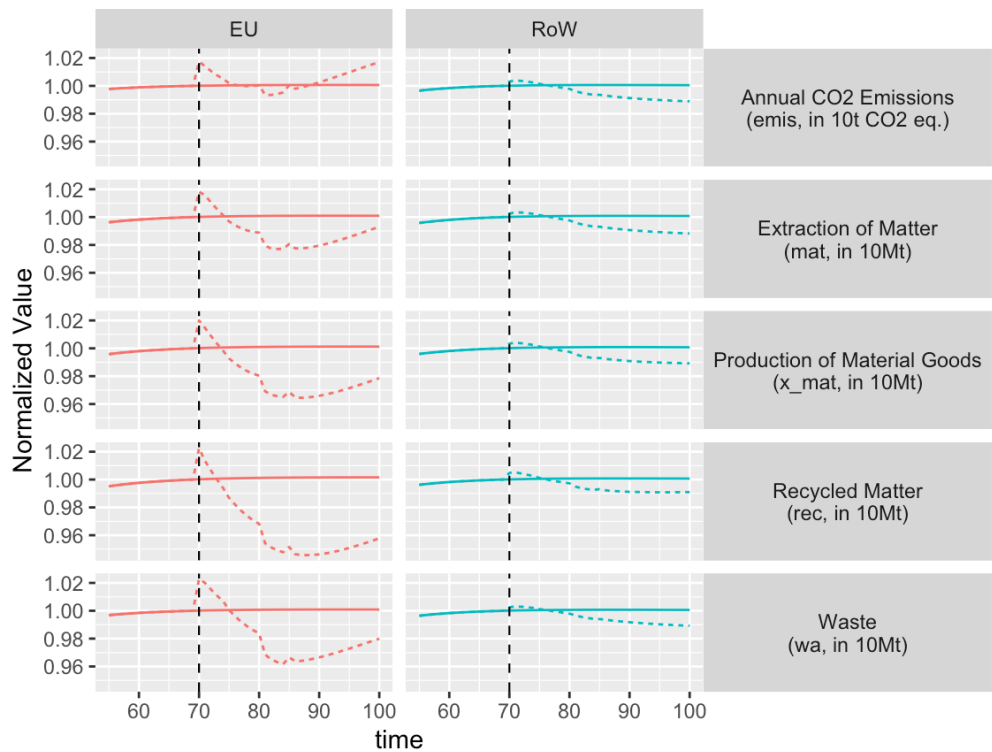
Selected Social Indicators. Vertical dashed line indicates shock time



Social Indicators

Scenario: Product Lifetime Extension

Selected Ecological Indicators. Vertical dashed line indicates shock time



Ecological Indicators

Government-led CE Transition



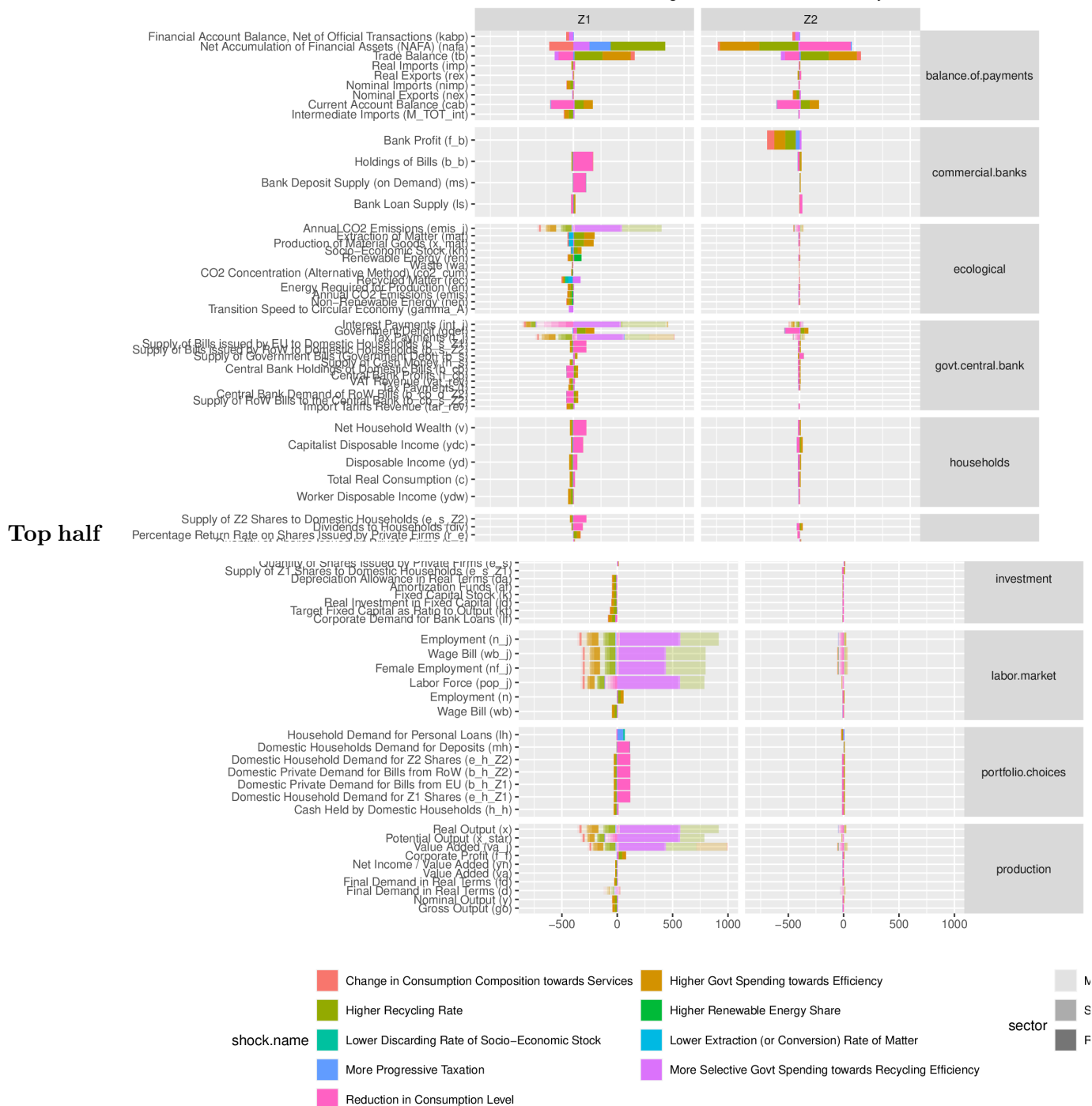
Scenario Summary: Change by Shock, Difference

Shock–Baseline Absolute Difference, Last Period by Shock
Only absolute values larger than 200 are shown.



Scenario Summary: Change by Shock, Percent

Shock–Baseline Percentage Difference, Last Period by Shock



Future Scenarios

Changes in Household Final Consumption Shares

- **Dietary transition:** reduced meat consumption (*Sector 7 → 8*)
- **Energy use transition:** (*Sector 31 → 32*)

Intermediate Production: **Shift from primary to secondary materials**

- Wood (*11 → 12*), Pulp (*13 → 14*), Plastics (*17 → 18*)
 - Metals (*26 → 27*), Glass (*21 → 22*), Cement (*24 → 25*)
 - Energy inputs (*31 → 32*)
 - Construction materials (*36 → 27*)
-

Future Work

- Make explicit the military sector (military Keynesianism) and migration as a dynamic feedback
- **Green value chain upgrading** from the NZIPL green tech dataset
- NZIPL Net-Zero Value Chain Explorer

Thank you!

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References